

TASK 3: GRAPH PATHFINDING AGENT

1. Executive Summary & Objective

Program: Progree AI Internship | Task: 3 (Graph Pathfinding)

Domain: Search Algorithms | Status: Completed | Evaluation: Optimality, Runtime, Expanded Nodes

This benchmark compares Dijkstra, A* with the Manhattan heuristic, and the bonus tabular Q-learning agent on four reproducible 4-connected grid mazes.

The deterministic algorithms provide an optimality baseline, while the learned agent demonstrates a model-free alternative. Every maze is generated with a fixed seed and verified reachable before evaluation.

Key outcomes

4	3	24	5
benchmark mazes	algorithms	automated tests	PNG visualizations

System architecture

Grid environment -> maze generator -> Dijkstra / A* / Q-learning -> metrics CSV and comparison visualizations

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2. Benchmark Metrics & Findings

Configuration	Algorithm	Solved	Steps	Runtime ms	Expanded	Cost
Easy (15×15, 10% density)	Dijkstra	Yes	28	1.86	200	28.0
Easy (15×15, 10% density)	A* (Manhattan)	Yes	28	3.49	169	28.0
Easy (15×15, 10% density)	Q-Learning (RL Agent)	Yes	28	960.52	29	28.0
Medium (20×20, 25% density)	Dijkstra	Yes	52	2.83	296	52.0
Medium (20×20, 25% density)	A* (Manhattan)	Yes	52	2.71	269	52.0
Medium (20×20, 25% density)	Q-Learning (RL Agent)	Yes	52	2909.74	53	52.0
Hard (25×25, 35% density)	Dijkstra	Yes	130	4.21	445	130.0
Hard (25×25, 35% density)	A* (Manhattan)	Yes	130	3.93	395	130.0
Hard (25×25, 35% density)	Q-Learning (RL Agent)	Yes	130	6153.29	131	130.0
Cul-de-Sac Trap (20×20, U-shaped)	Dijkstra	Yes	38	5.73	353	38.0
Cul-de-Sac Trap (20×20, U-shaped)	A* (Manhattan)	Yes	38	3.21	295	38.0
Cul-de-Sac Trap (20×20, U-shaped)	Q-Learning (RL Agent)	Yes	38	1314.95	39	38.0

Key findings

Dijkstra and A* produce equal optimal path costs on every solvable maze. A* expands no more nodes than Dijkstra in these configurations because the Manhattan heuristic guides the frontier toward the goal. Q-learning is stochastic and its runtime includes training, so its timing is not directly comparable to one-pass deterministic search.

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3. Implementation & Reproduction

Environment

Bounded grid state, obstacle handling, 4-connected movement, and admissible distance heuristics.

Search algorithms

Dijkstra provides the optimal uniform-cost baseline. A* adds the Manhattan heuristic. Tabular Q-learning trains a policy and extracts a greedy route.

Outputs

The benchmark exports a metrics CSV, four per-maze comparison panels, and one aggregate performance chart.

Validation

The repository test suite covers grid behavior, heuristic validity, path correctness, optimality, unsolvable grids, and all four benchmark configurations.

Reproduce the benchmark

```
.\.venv\Scripts\python.exe benchmark.py
```

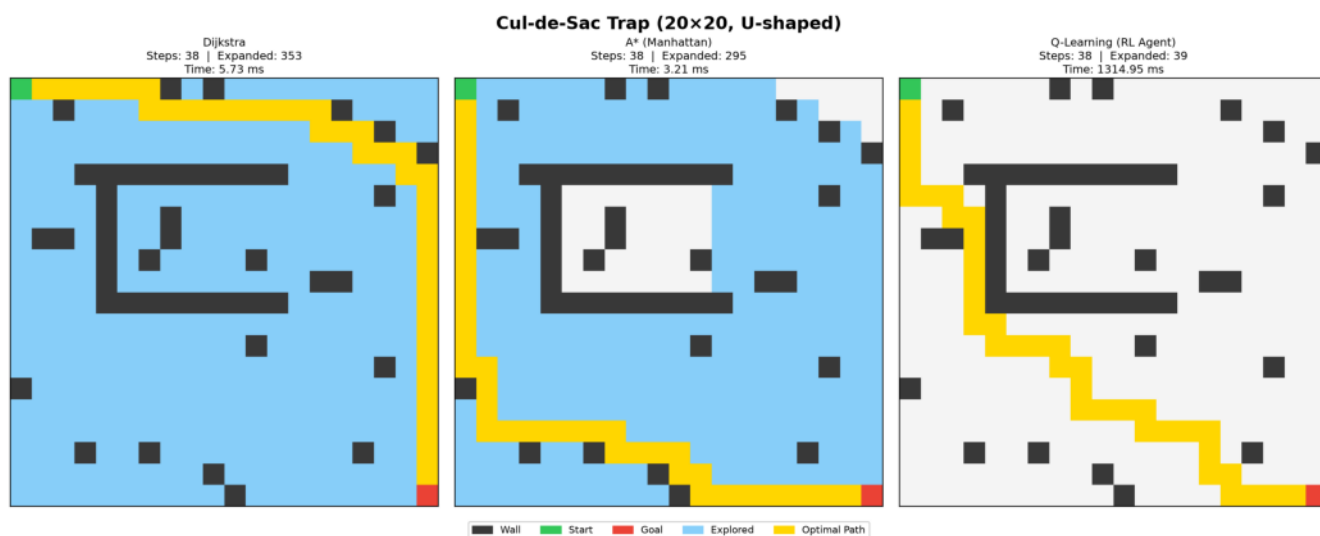
The generated Markdown report includes the same five figures shown on the following pages.

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4. Visual Results

Cul-de-Sac trap comparison



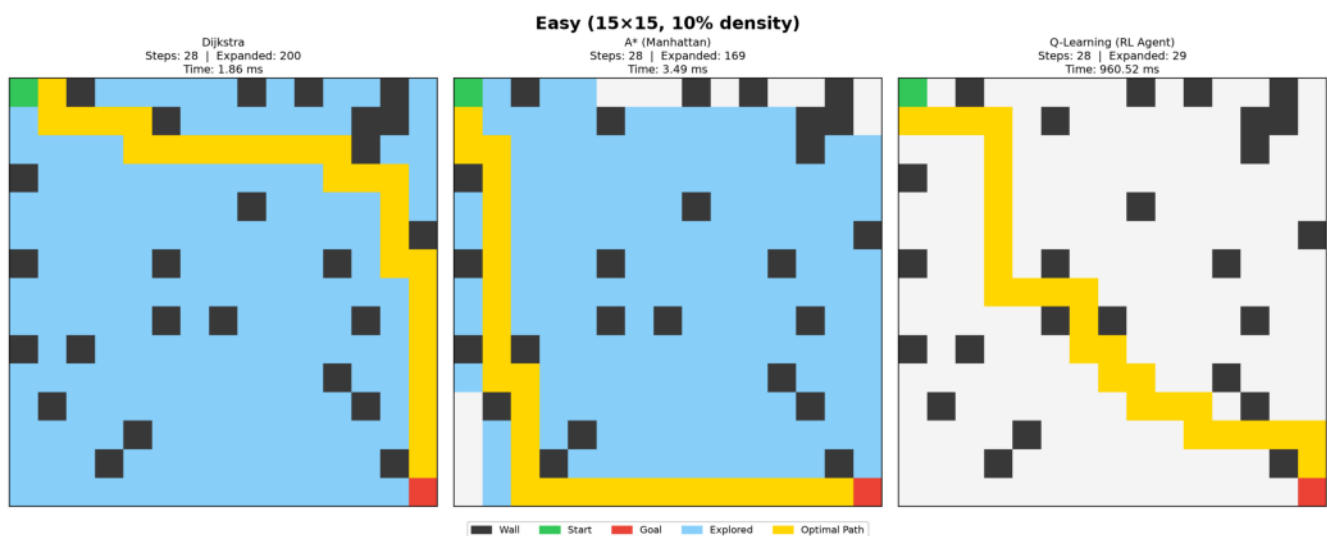
Generated from the fixed-seed benchmark run. See REPORT.md for the corresponding interpretation and metric definitions.

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4. Visual Results

Easy maze comparison



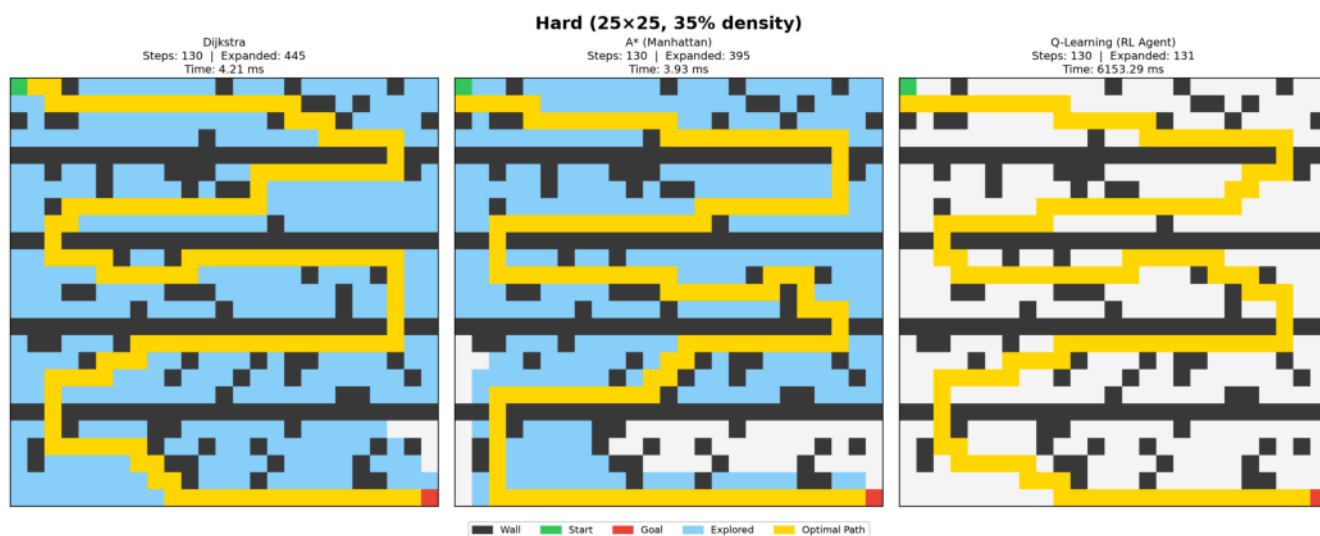
Generated from the fixed-seed benchmark run. See REPORT.md for the corresponding interpretation and metric definitions.

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4. Visual Results

Hard maze comparison



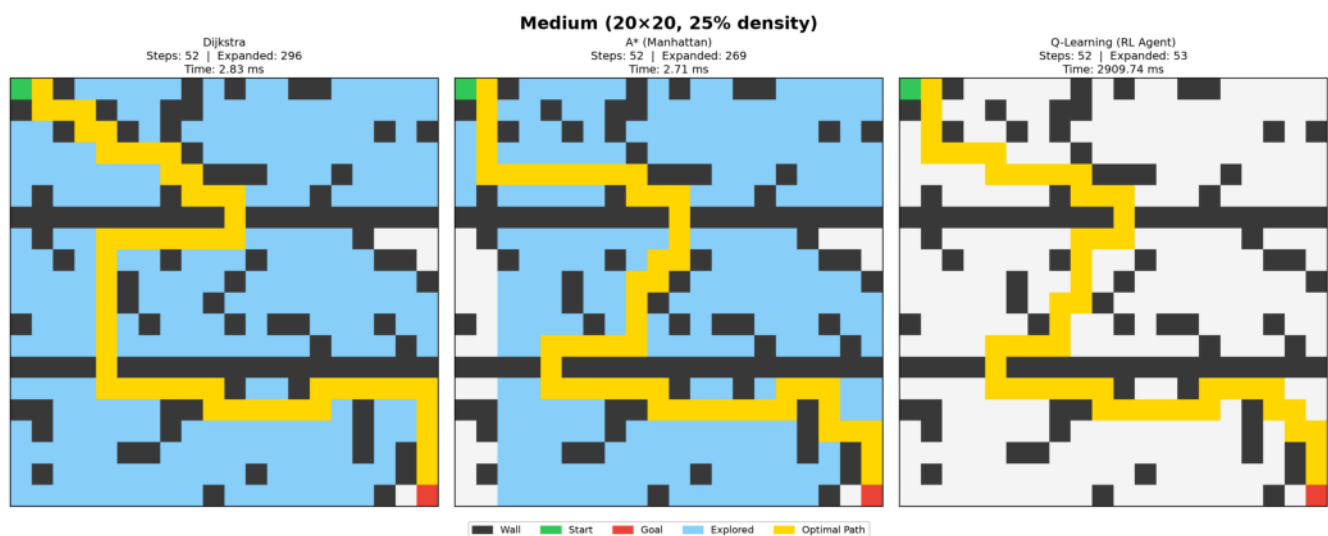
Generated from the fixed-seed benchmark run. See REPORT.md for the corresponding interpretation and metric definitions.

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4. Visual Results

Medium maze comparison



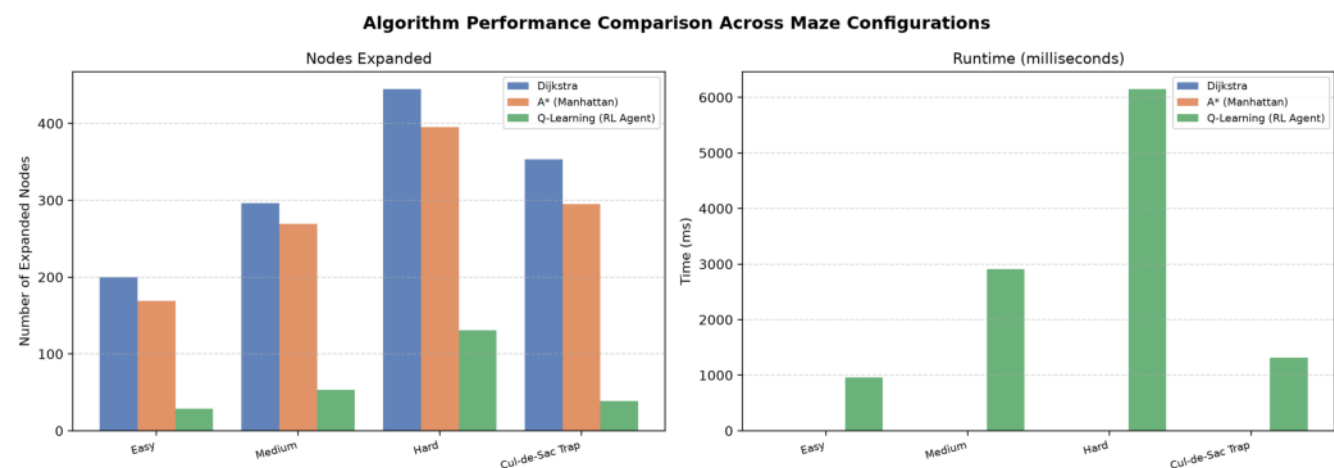
Generated from the fixed-seed benchmark run. See REPORT.md for the corresponding interpretation and metric definitions.

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4. Visual Results

Aggregate metrics comparison



Generated from the fixed-seed benchmark run. See REPORT.md for the corresponding interpretation and metric definitions.